Year 11 Higher

REVISION PACK

Handling Data 5

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Handling Data 5 · Averages from frequency tables (Lesson 9.1)

Q	Working	Answer
grouped data		
1.1	**(a)** **(b)** $(40+1)\div 2 = 20.5\text{th}$ $\text{add: } 4,\ 13,\ 28$ **(c)** $5\times 4+15\times 9+25\times 15+35\times 8+45\times 4 = 990$ $990\div 40 = 24.75$	**(a)** $20 < t \leq 30$ **(b)** $20 < t \leq 30$ **(c) 24.75 minutes**
1.2	**(a)** **(b)** $(50+1)\div 2 = 25.5\text{th}$ $\text{add: } 6,\ 20,\ 40$ **(c)** $10\times 6+30\times 14+50\times 20+70\times 7+90\times 3 = 2240$ $2240\div 50 = 44.8$	**(a)** $40 < m \leq 60$ **(b)** $40 < m \leq 60$ **(c) 44.8 g**
1.3	**(a)** **(b)** $(40+1)\div 2 = 20.5\text{th}$ $\text{add: } 3,\ 14,\ 26$ **(c)** $2.5\times 3+7.5\times 11+12.5\times 12+17.5\times 10+22.5\times 4 = 505$ $505\div 40 = 12.62$	**(a)** $10 < t \leq 15$ **(b)** $10 < t \leq 15$ **(c) 12.62 minutes**
1.4	**(a)** **(b)** $(40+1)\div 2 = 20.5\text{th}$ $\text{add: } 2,\ 10,\ 24$ **(c)** $5\times 2+15\times 8+25\times 14+35\times 11+45\times 5 = 1090$ $1090\div 40 = 27.25$	**(a)** $20 < t \leq 30$ **(b)** $20 < t \leq 30$ **(c) 27.25 minutes**

1.5	**(a)** **(b)** $(40+1)\div 2 = 20.5\text{th}$ $\text{add: } 7,\ 20,\ 29$ **(c)** $2\times 7+6\times 13+10\times 9+14\times 6+18\times 5=356$ $356\div 40=8.9$	**(a)** $4<t\leq 8$ **(b)** $8<t\leq 12$ **(c) 8.9 seconds**
1.6	**(a)** **(b)** $(50+1)\div 2 = 25.5\text{th}$ $\text{add: } 5,\ 15,\ 30$ **(c)** $5\times 5+15\times 10+25\times 15+35\times 12+45\times 8=1330$ $1330\div 50=26.6$	**(a)** $20<a\leq 30$ **(b)** $20<a\leq 30$ **(c) 26.6 years**
ungrouped data		
1.7	**(a)** $0\times 3+1\times 6+2\times 5+3\times 4+4\times 2=36$ $36\div 20=1.8$ **(b)** $(20+1)\div 2 = 10.5\text{th}$ $\text{add: } 3,\ 9,\ 14,\ 18,\ 20$	**(a) 1.8** **(b) 2**
1.8	**(a)** $1\times 4+2\times 7+3\times 5+4\times 3+5\times 1=50$ $50\div 20=2.5$ **(b)** $(20+1)\div 2 = 10.5\text{th}$ $\text{add: } 4,\ 11,\ 16,\ 19,\ 20$	**(a) 2.5** **(b) 2**
1.9	**(a)** $0\times 5+1\times 8+2\times 4+3\times 3=25$ $25\div 20=1.25$ **(b)** $(20+1)\div 2 = 10.5\text{th}$ $\text{add: } 5,\ 13,\ 17,\ 20$	**(a) 1.25** **(b) 1**
1.10	**(a)** $0\times 2+1\times 5+2\times 8+3\times 3+4\times 2=38$ $38\div 20=1.9$ **(b)** $(20+1)\div 2 = 10.5\text{th}$ $\text{add: } 2,\ 7,\ 15,\ 18,\ 20$	**(a) 1.9** **(b) 2**

1.11	**(a)**	$1 \times 6 + 2 \times 9 + 3 \times 3 + 4 \times 2 = 41$ $41 \div 20 = 2.05$	**(a) 2.05** **(b) 2**
	(b)	$(20 + 1) \div 2 = 10.5\text{th}$ $\text{add: } 6,\ 15,\ 18,\ 20$	
1.12	**(a)**	$0 \times 1 + 1 \times 4 + 2 \times 6 + 3 \times 5 + 4 \times 3 + 5 \times 1 = 48$ $48 \div 20 = 2.4$	**(a) 2.4** **(b) 2**
	(b)	$(20 + 1) \div 2 = 10.5\text{th}$ $\text{add: } 1,\ 5,\ 11,\ 16,\ 19,\ 20$	